

GABLS1: 64³ resolution study

GABLS1 32-cubed to 64-cubed resolution sensitivity | 23 September 2026. Matched nine-hour WENO9 cases use a 400 m cube, seed 123, filtered rough-wall drag, and paired initial perturbations. The gamma=2 scheme-native SLD uses a 300 s filter. Fixed 1 m LES medians are intercomparison references, not observations. Shear below is computed on common 12.5 m physical-height intervals after interpolating both u and v to coarse-grid centers; native shear is shown separately.

During 8-9 h, the 64-cubed one-face SLD shear at 12.5, 25, and 37.5 m is 0.10144, 0.05600, and 0.03951 per second. The two-face values are 0.10348, 0.05945, and 0.03972. The fixed 1 m median is 0.10038, 0.06666, and 0.05607. The second face improves 25 m agreement modestly but does not repair the 37.5 m deficit. Filtered no closure at 64-cubed nearly matches the first reference shear, 0.09953, yet is low at the next two heights. Two grids reveal sensitivity, not established convergence.

Peak resolved w-squared in 8-9 h rises from 0.05588 square meters per second squared for 32-cubed one-face SLD to 0.09688 for 64-cubed one-face and 0.08933 for 64-cubed two-face, near the 64-cubed no-closure value 0.10113. At 18.75 m, interpolated w-cubed changes from +0.000752 to -0.000777 and -0.001543 cubic meters per second cubed respectively. The fixed 1 m archive lacks a w-cubed median; this is not a fidelity ranking. Both time windows appear in the figures.

At the physical face 12.5 m, 32-cubed one-face and 64-cubed two-face have different transport partitions even though support height matches. Mean 64-cubed two-face viscosity is 0.09525 square meters per second, resolved u-w is -0.04946, SGS u-w is -0.00776, and separate WENO correction is -0.01178 square meters per second squared. The corresponding 32-cubed viscosity is 0.52303. One-face at 64-cubed acts at only 6.25 m, so its support height is different. All comparisons retain the same wall law and physical forcing.

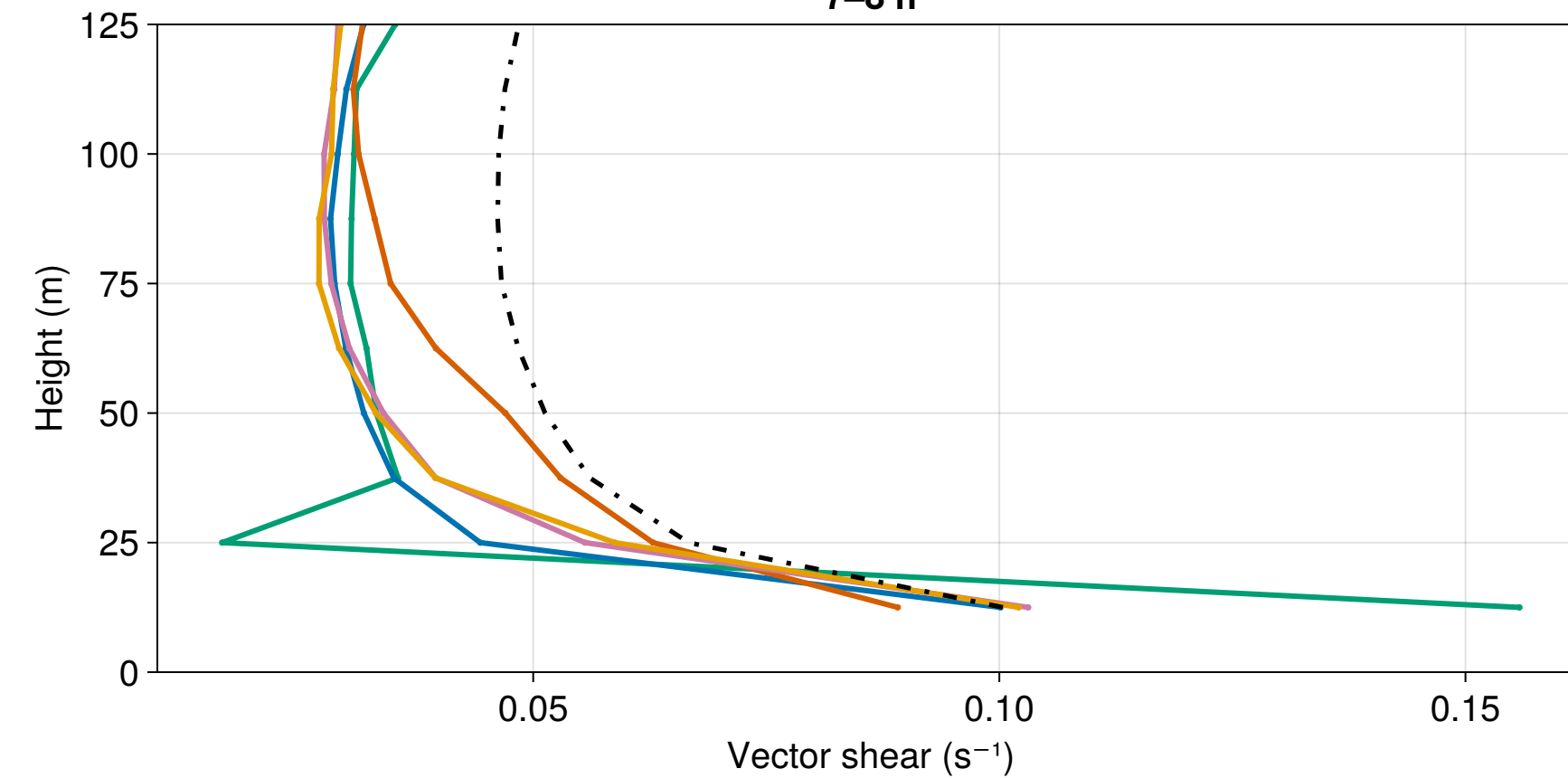
Validation: immutable 64-cubed source-manifest SHA-256

19f41d02f5bd249fe32a5edf2d8956e9886b1f58347a98907ec83b1c23d0fbd7. H100 gate 7627 passed CUDA parity and 15/15, 32/32, 38/38 case checks. Science array 7640 tasks 1-3 ended with durable exit zero and CASE_DONE at 32400 s. Independent frozen Julia audits verified source, paired initial state, native 64/65 levels, exact schedules, finite fields, evolving wall filter, density-consistent surface flux, and SLD flux identities. Both SLD cases passed 55 saved local 1/L and similarity records through nine hours. Compact exports and scripts accompany this chapter. A paired 128-cubed extension follows; current SLD supports at most two faces, so it cannot retain 12.5 m support at 128-cubed without additional code.

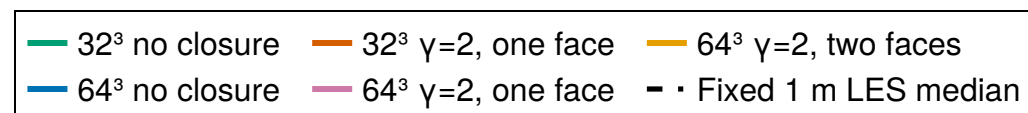
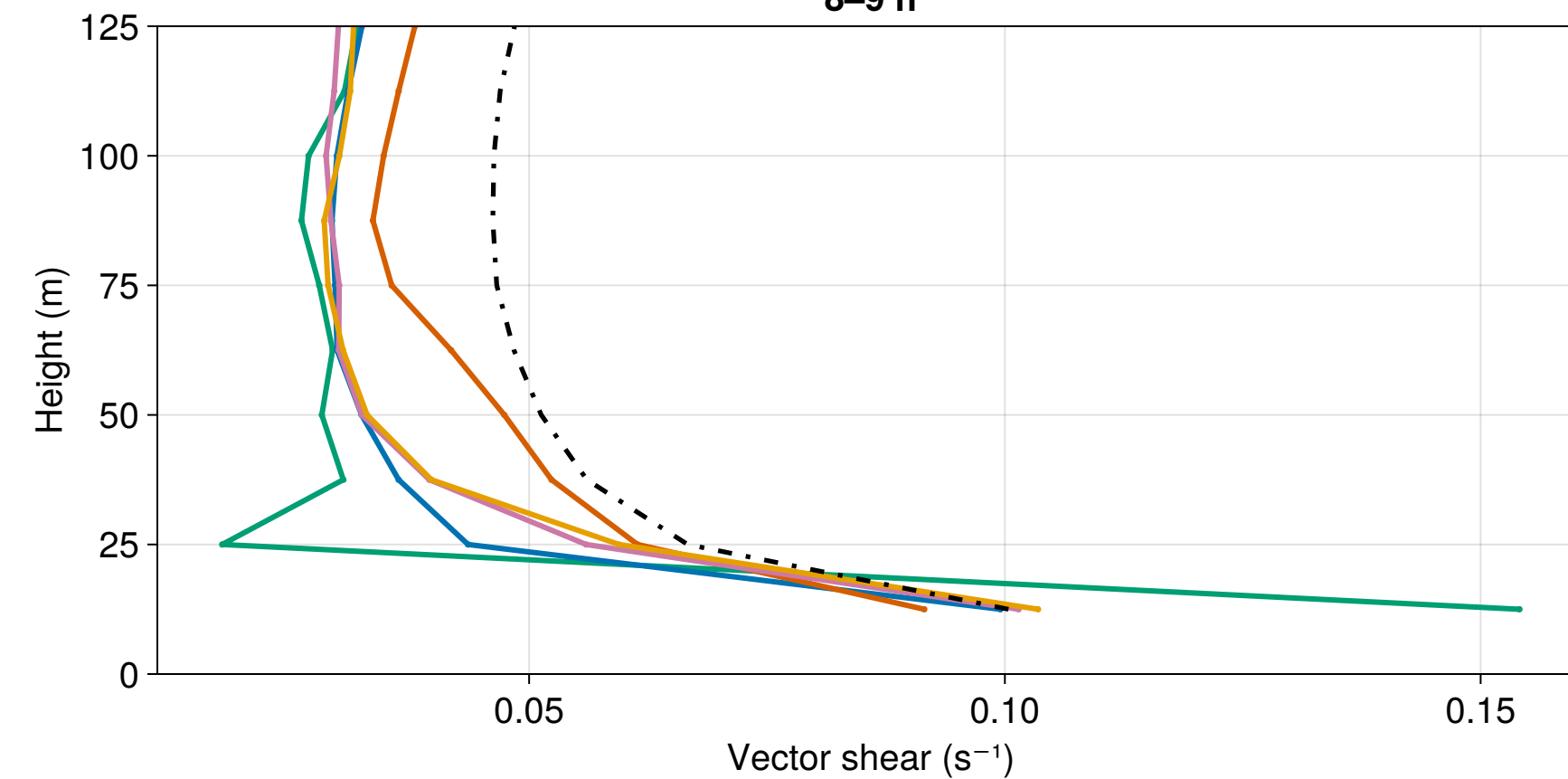
Vector shear at matched physical heights

u and v interpolated to 32^3 cell centers (6.25, 18.75, ... m), then differenced

7–8 h

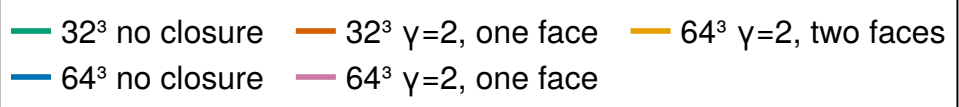
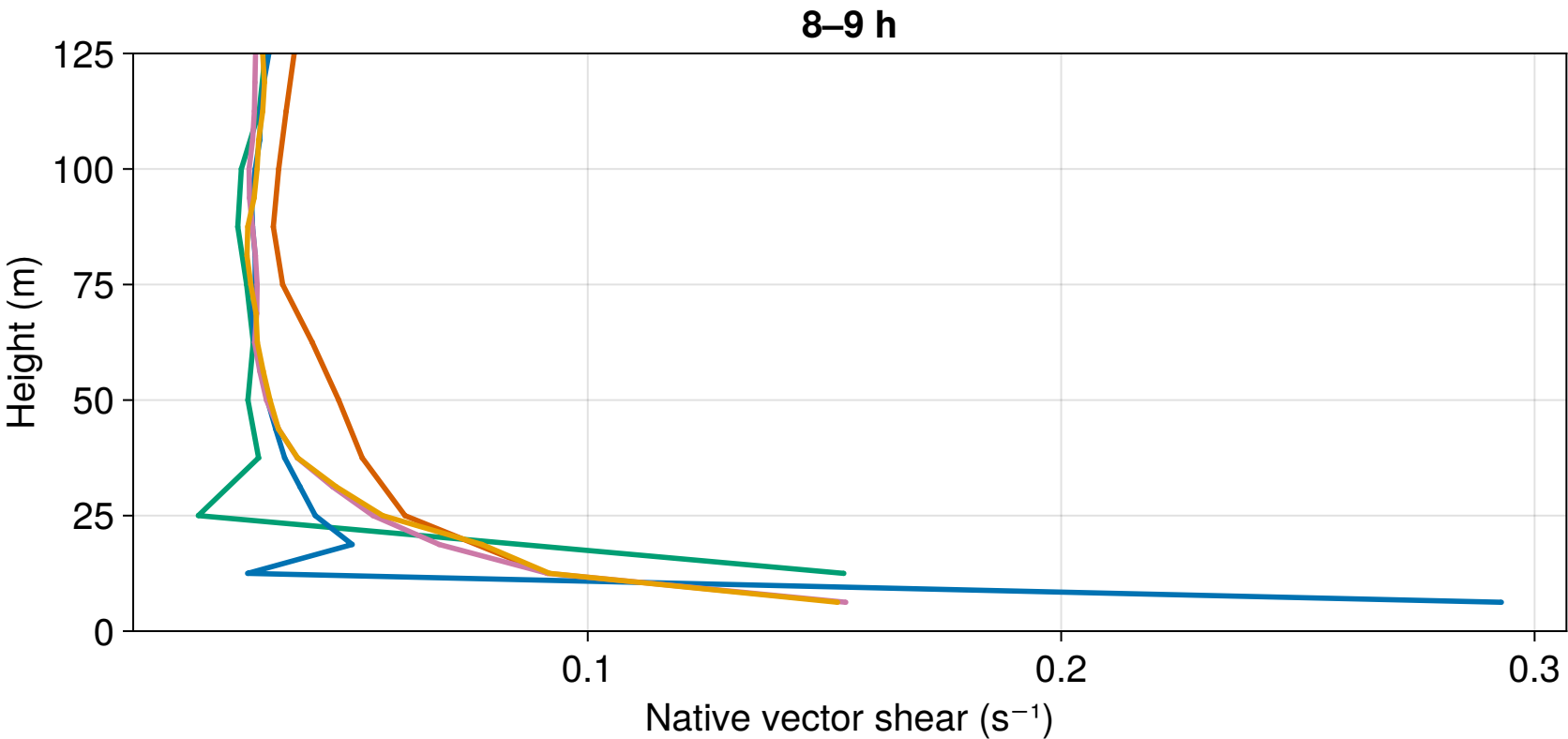
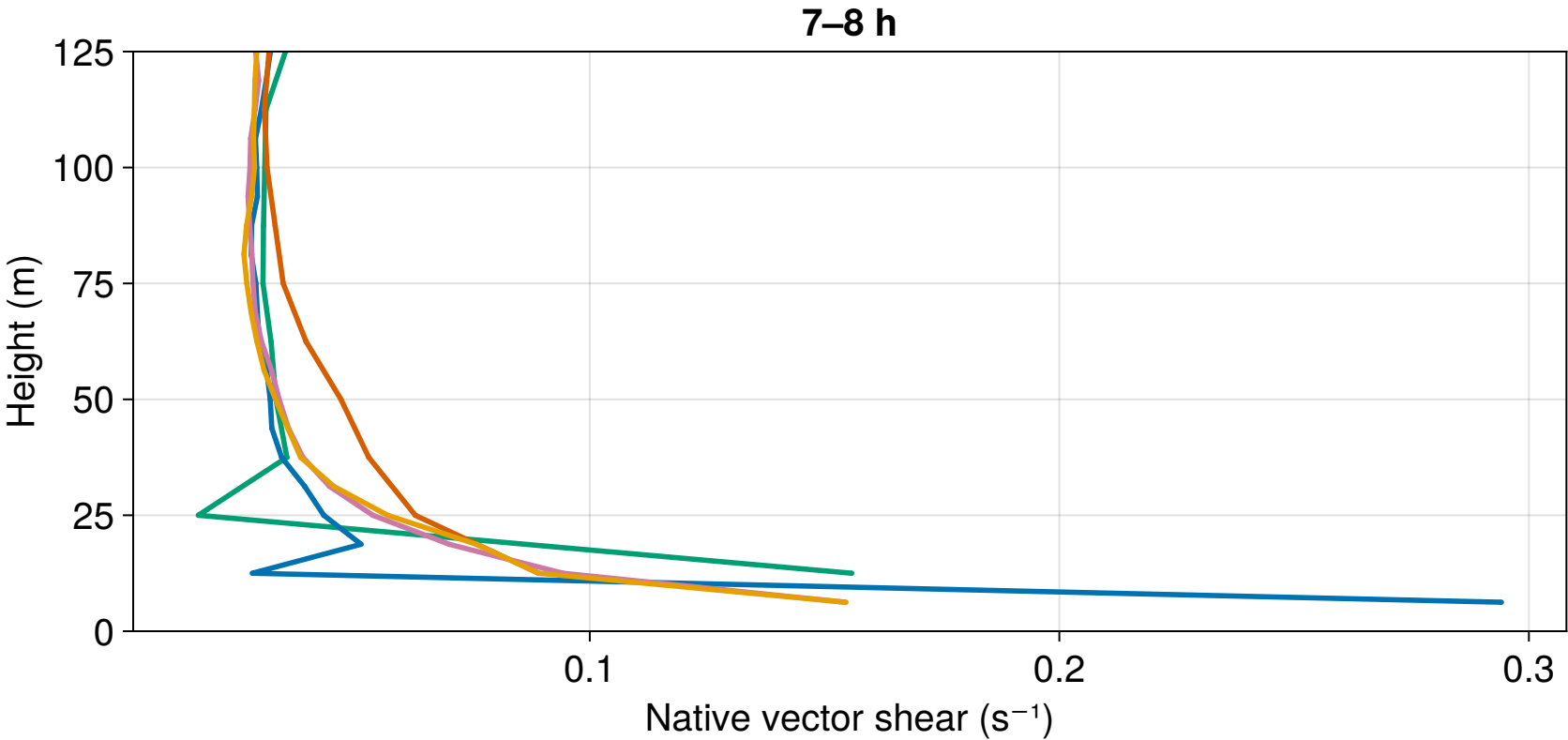


8–9 h



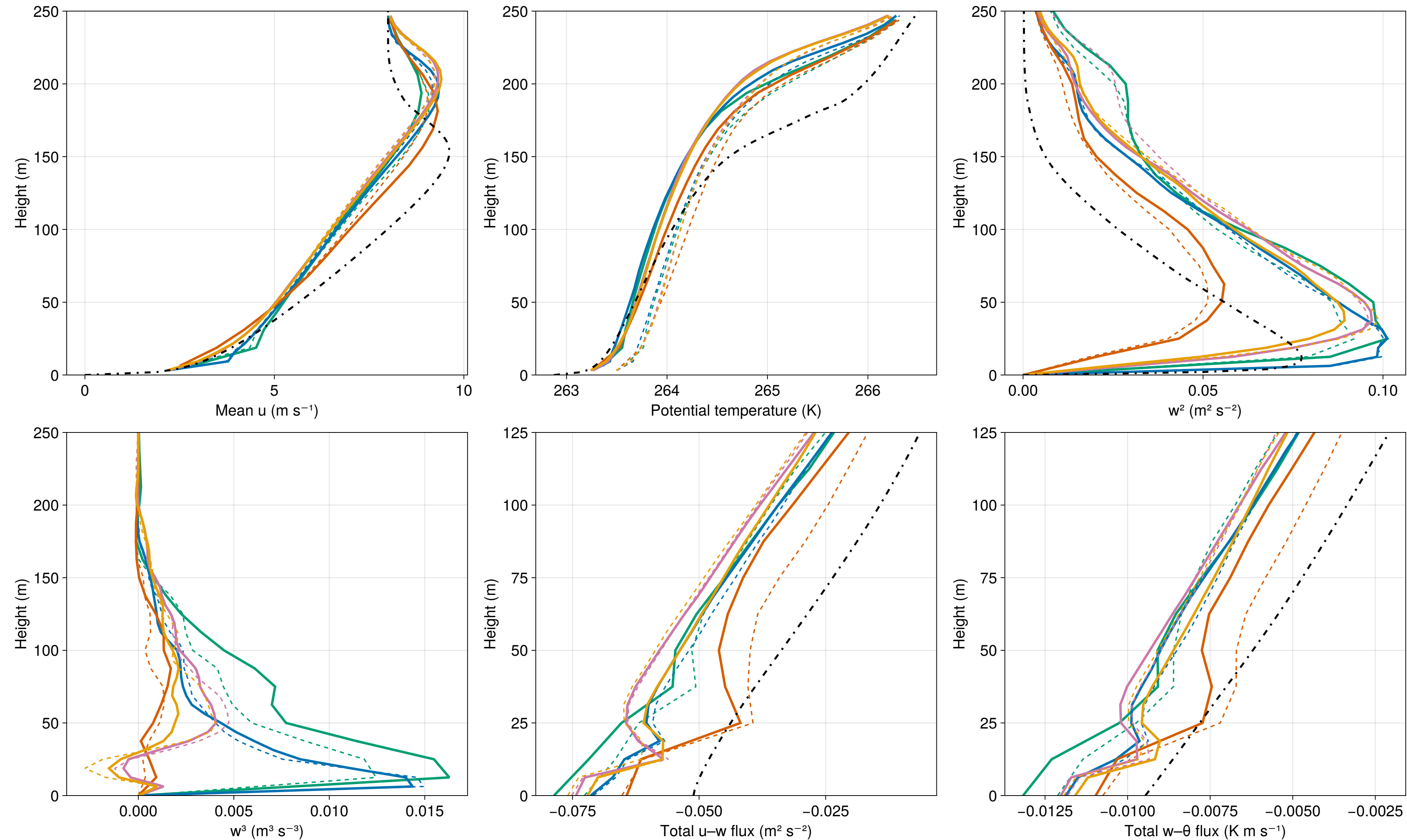
Native-grid vector shear

64³ shear spans 6.25 m; 32³ shear spans 12.5 m. Compare with matched-height figure.



GABLS1 $32^3 \rightarrow 64^3$: matched stability-corrected SLD

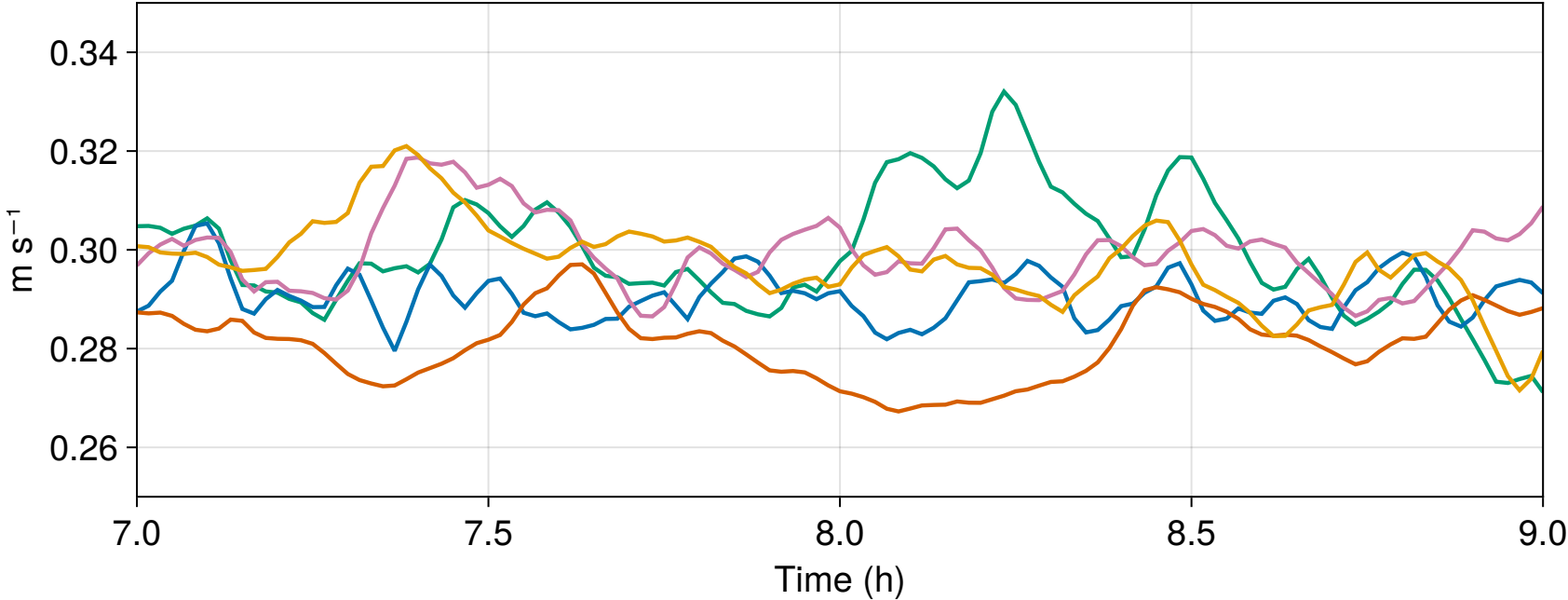
400 m cube · WENO9 · $\gamma=2$ · 300 s wall/SLD filters · dashed 7–8 h, solid 8–9 h



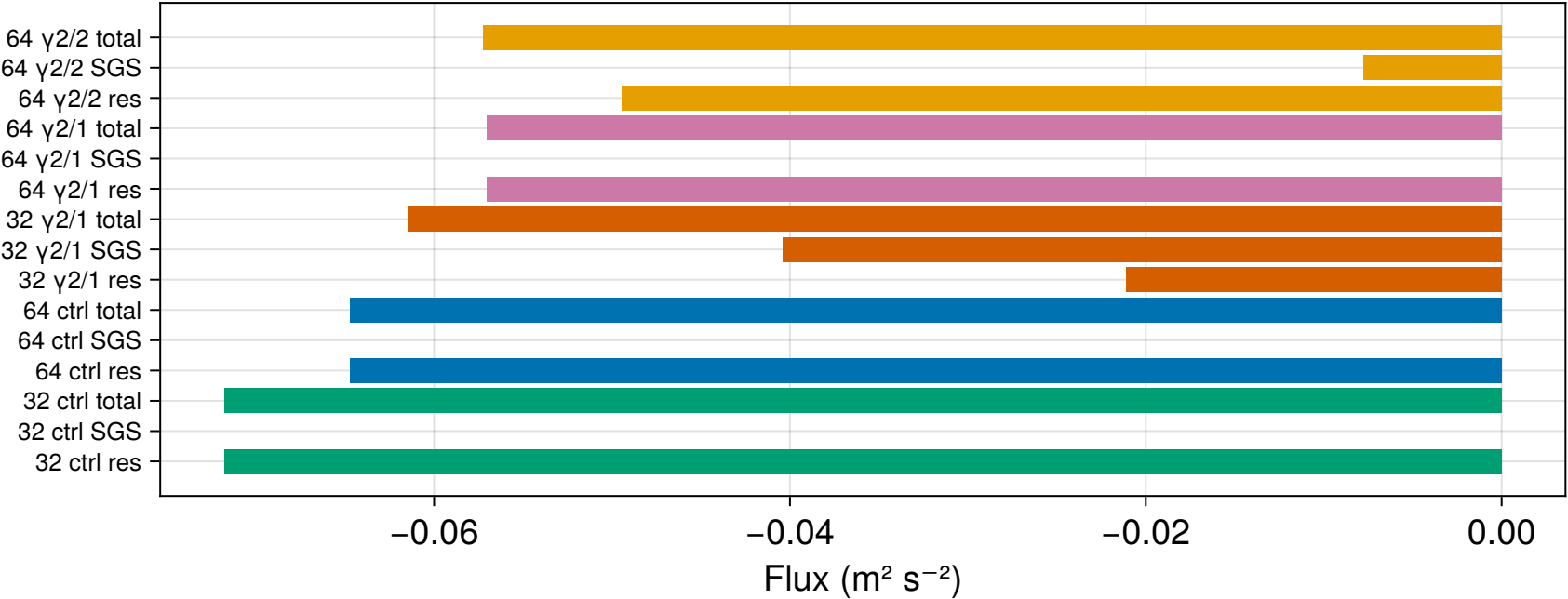
— 32^3 no closure — 32^3 $\gamma=2$, one face — 64^3 $\gamma=2$, two faces
— 64^3 no closure — 64^3 $\gamma=2$, one face - - Fixed 1 m LES median

Wall response and vertical-transport partition

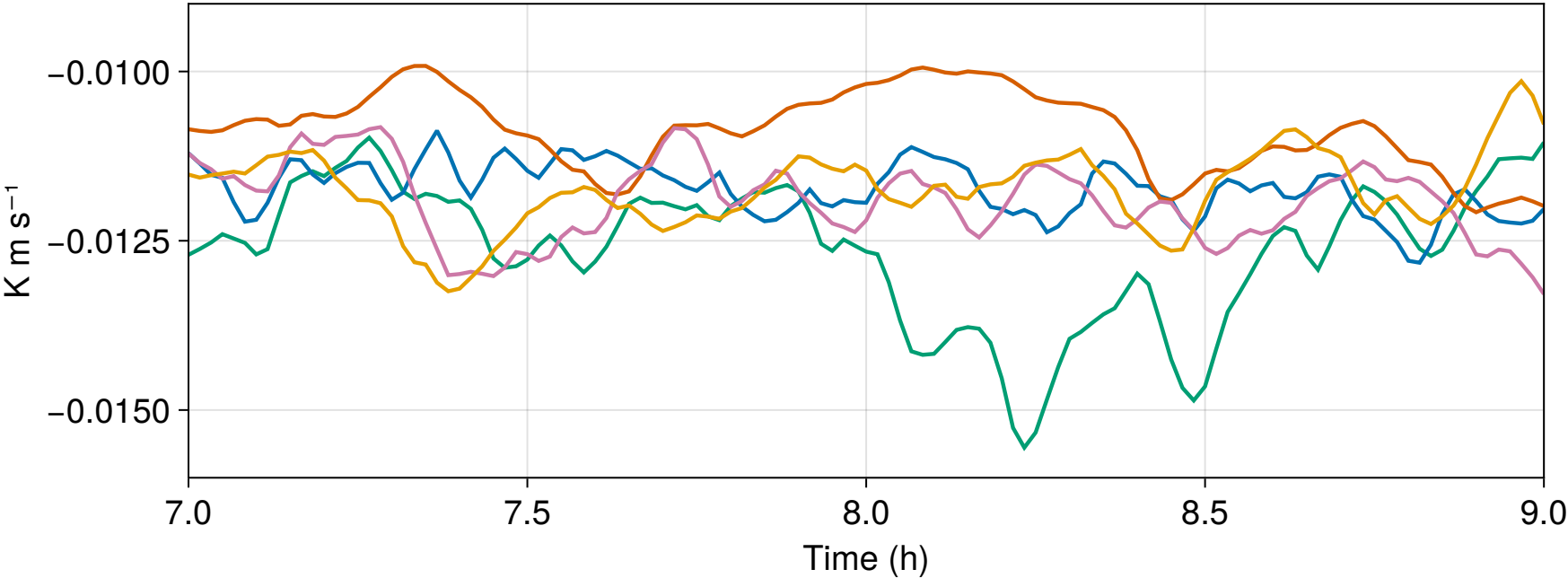
Friction velocity



u-w at z=12.5 m



Surface heat flux



w-θ at z=12.5 m

